

Measures of spread continued



Overview

Quick review of measures of spread, z-scores, and percentile ranks

Continuation of measures of variability

- Percentiles
- Box plot

Relationships between two variables

- Scatter plots and correlation

Announcement: homework 2

Homework 2 is due on Gradescope
on Sunday, September 20th at 11pm

[library\(SDS1000\)](#)

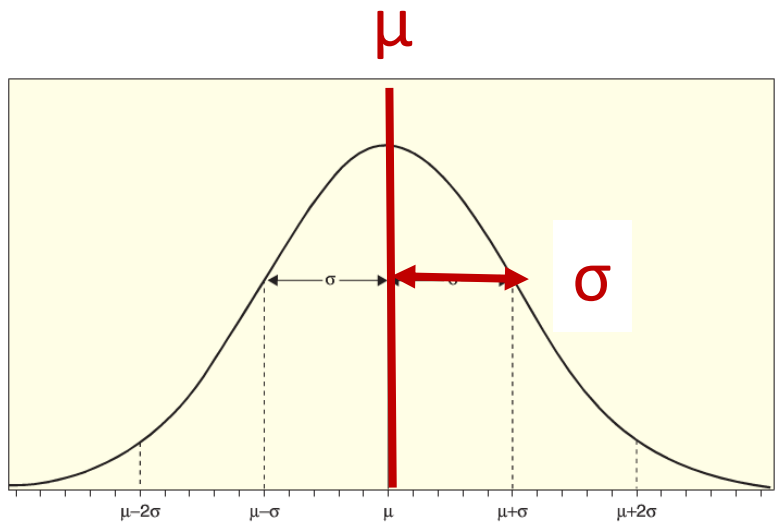
[goto_homework\(2\)](#)

Note: you will also need to fill out a **worksheet survey** on sampling data from the Gettysburg address as part of the reflection



Keep attending the practice sessions!

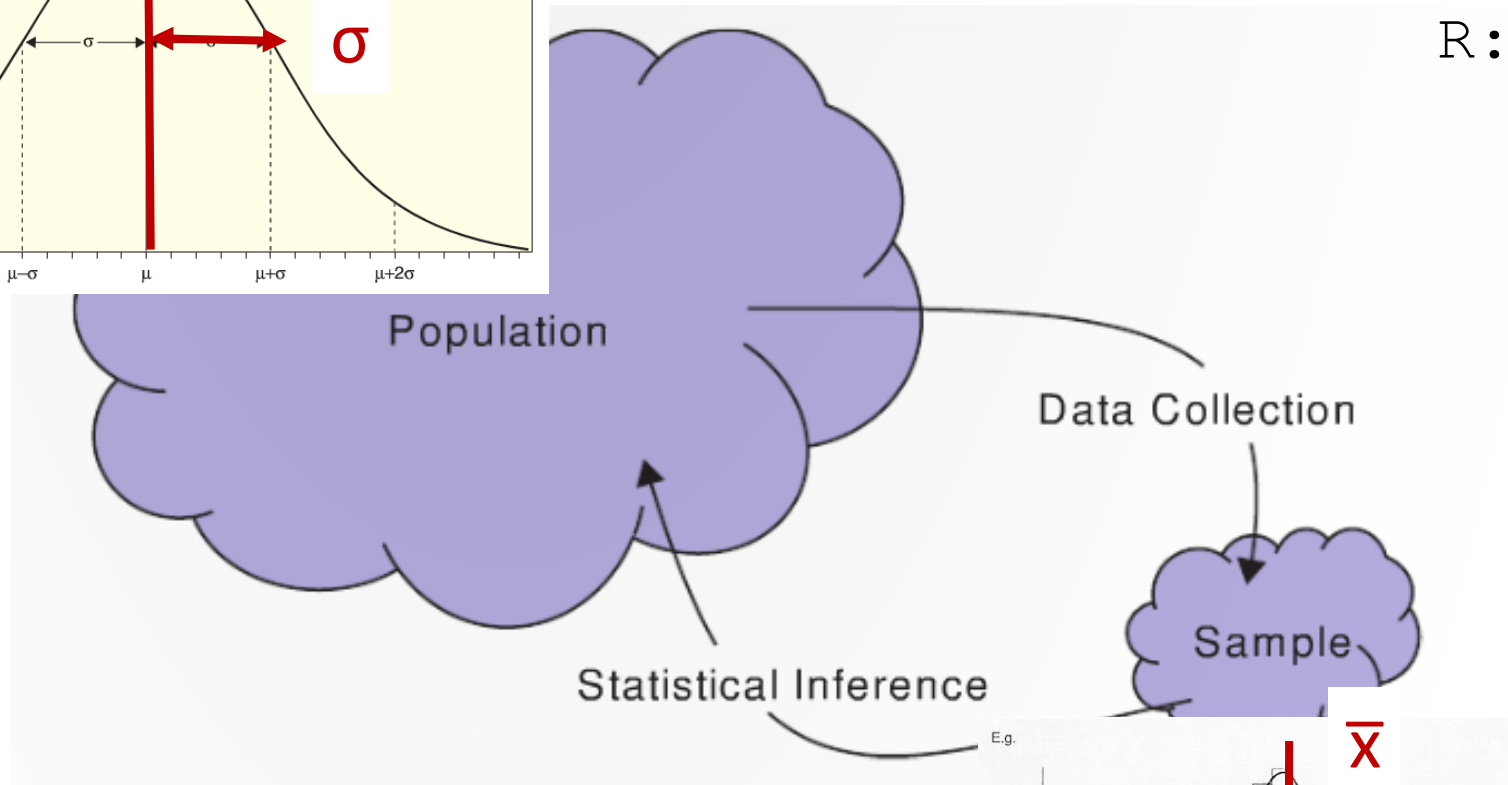
Review measures of spread...



Parameters

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

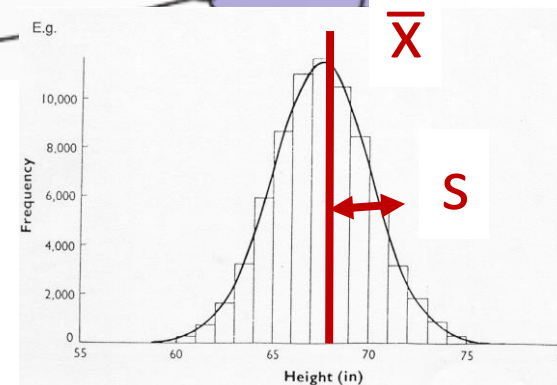
R: `mean(x)`



$$s = \sqrt{\frac{1}{(n-1)} \sum_{i=1}^n (x_i - \bar{x})^2}$$

R: `sd(x)` and `var(x)`

statistics



Review: z-scores

The z-scores tells how many standard deviations a value is from the mean

$$\text{z-score}(x_i) = \frac{x_i - \bar{x}}{s}$$



Which statistic is most impressive?

Z-score FGPct = 0.868

Z- score Points = 2.698

Z-score Assists = 1.965

Z-score Steals = 1.771

This is LeBron's best statistic (relative to his peers)
But should we be impressed with this?

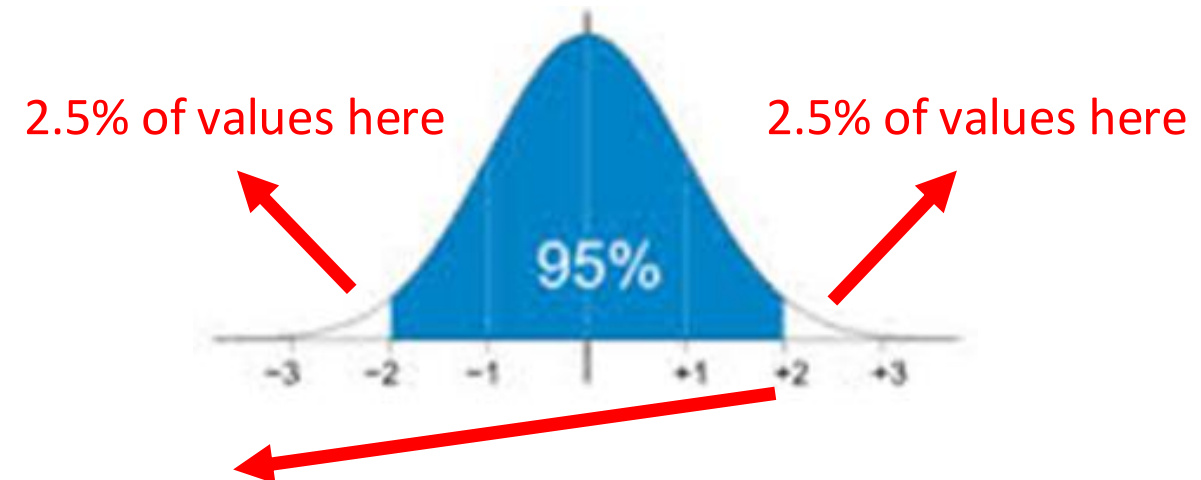
- i.e., maybe z-scores of 2.7 are common?

The normal pillow

Question: What percent of the pillow's fluff is within ± 2 standard deviations from the mean?



Question: If the values are normally distributed, how frequently should we expect a randomly selected z-score to be more than 2?



97.5% of values are less than 2

Review: Percentile rank

The **percentile rank** is the percentage of values in a data set that are less than or equal to a value **x**

We can use `ptail(x, vec_values)` to calculate the percentile rank (returned as a proportion)

For example, suppose we had:

```
ping_data <- c(6, 12, 45, 59, 158)
```

What is the percentile rank of 15?

Answer: 40% `ptail(15, ping_data)` # returns 0.4

We can also get the proportion of values greater or equal to x using:

```
ptail(x, vec_values, lower.tail = FALSE)
```

Percentiles

Percentiles

The **P^{th} percentile** is the value (v) which is greater than P percent of the data

- i.e., $p\%$ of the data is less than v
- (inverse of the percentile rank)

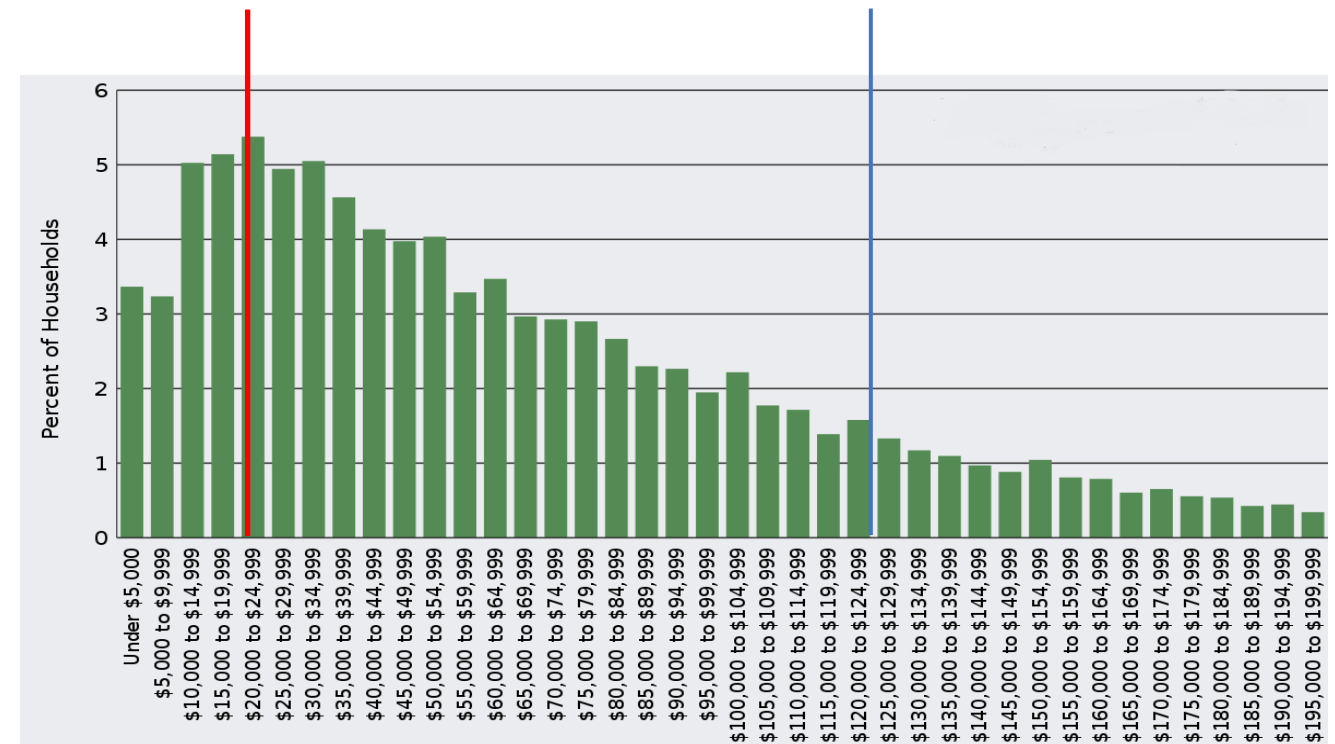
For the US income distribution what are the 20th and 80th percentiles?

20th percentile = \$21,430

80th percentile = \$112,254

Note of caution: there is not just one way to define a percentile

- Some definitions always use values in the data sample
- Other definitions interpolate between data points



R: `quantile(v, .95)`

Age of best actors

The **Academy Awards**, give out **Oscars** to the best actor and best actress each year

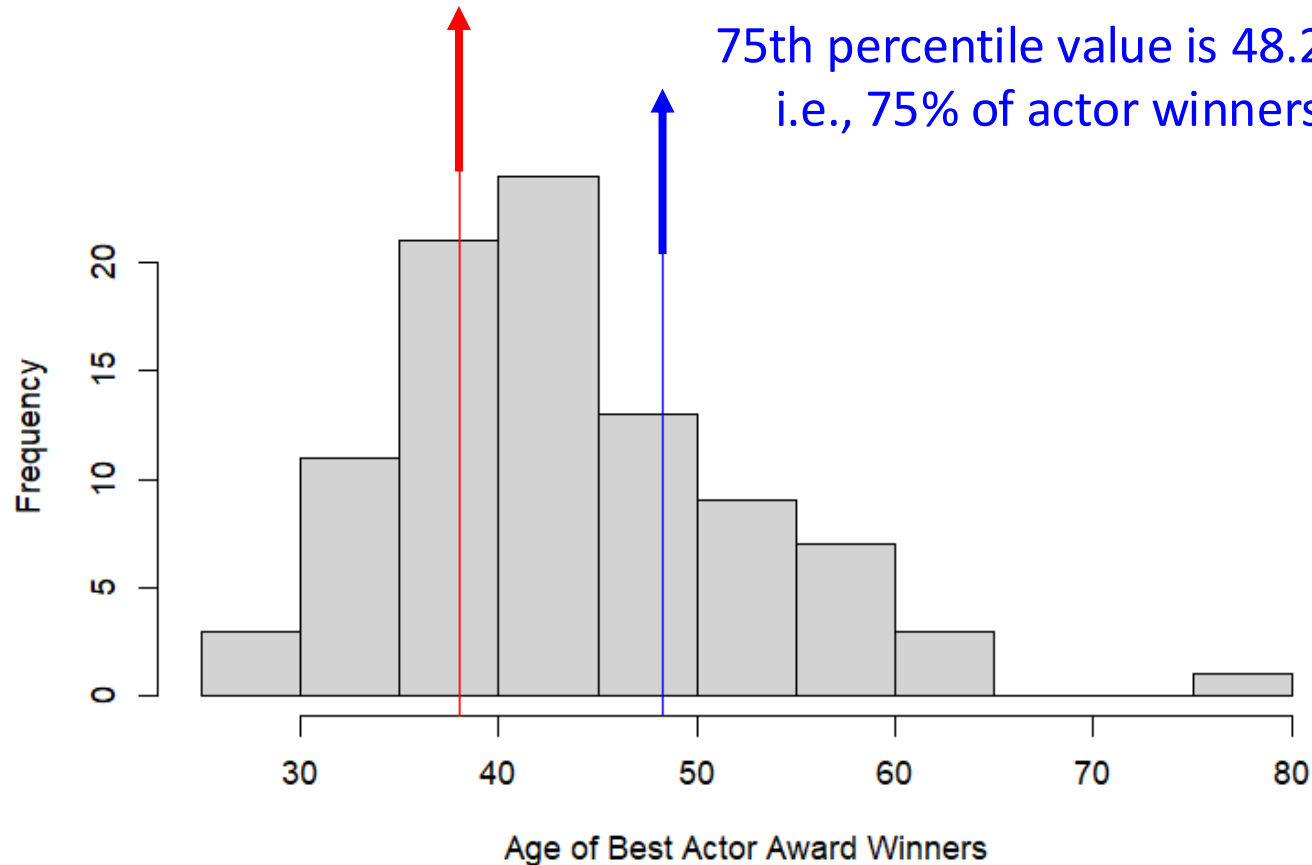
Let's examine the age of these Oscar winners from the years 1929 to 2012



Age of best actors

25th percentile value is 38

i.e., 25% of actor winners were 38 or younger



75th percentile value is 48.25

i.e., 75% of actor winners were 48.25 or younger

Middle 50% of winners
(25th to 75th percentiles)

In the age range:

- 38 to 48.25 years old

Five Number Summary

Five Number Summary = (minimum, Q_1 , median, Q_3 , maximum)

Q_1 = 25th percentile (also called 1st quartile)

Q_3 = 75th percentile (also called 3rd quartile)

Roughly divides the data into fourths

Calculating quartiles “by hand”

Our sorted ping data is: 4 6 10 12 45 59 158

1. Calculate the median as the middle of the sorted data

2. For all values less than the median, calculate the median of these values, which will give you Q_1

3. For all values greater than the median, calculate the median of these values, which will give you Q_3



Note of caution (again): there is not just one way define a percentile

- Some definitions always use values in the data sample
- Other definitions interpolate between data points

Range and Interquartile Range

Other measures of spread are:

Range = maximum – minimum

Interquartile range (IQR) = $Q_3 - Q_1$

Detecting of outliers

As a rule of thumb, we call a data value an **outlier** if it is:

Smaller than: $Q_1 - 1.5 * IQR$

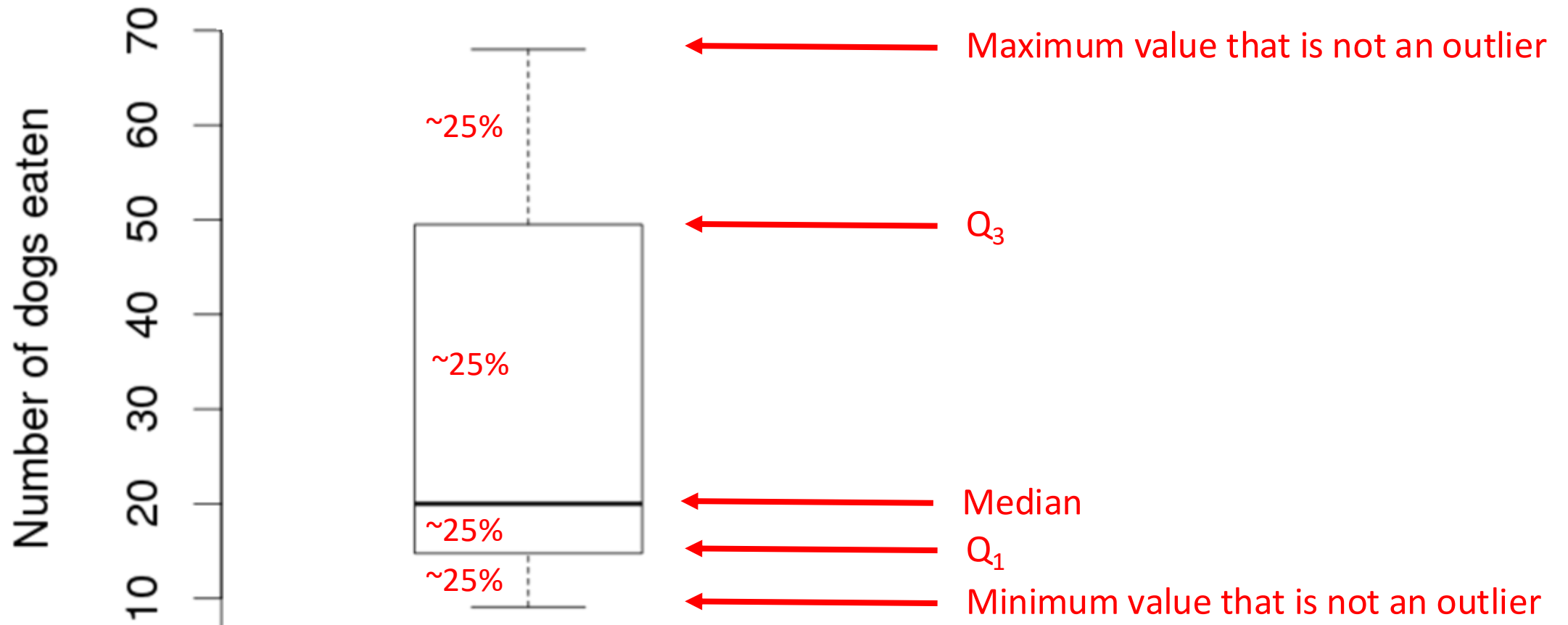
Larger than: $Q_3 + 1.5 * IQR$

Box plots

A **box plot** is a graphical display of the five-number summary and consists of:

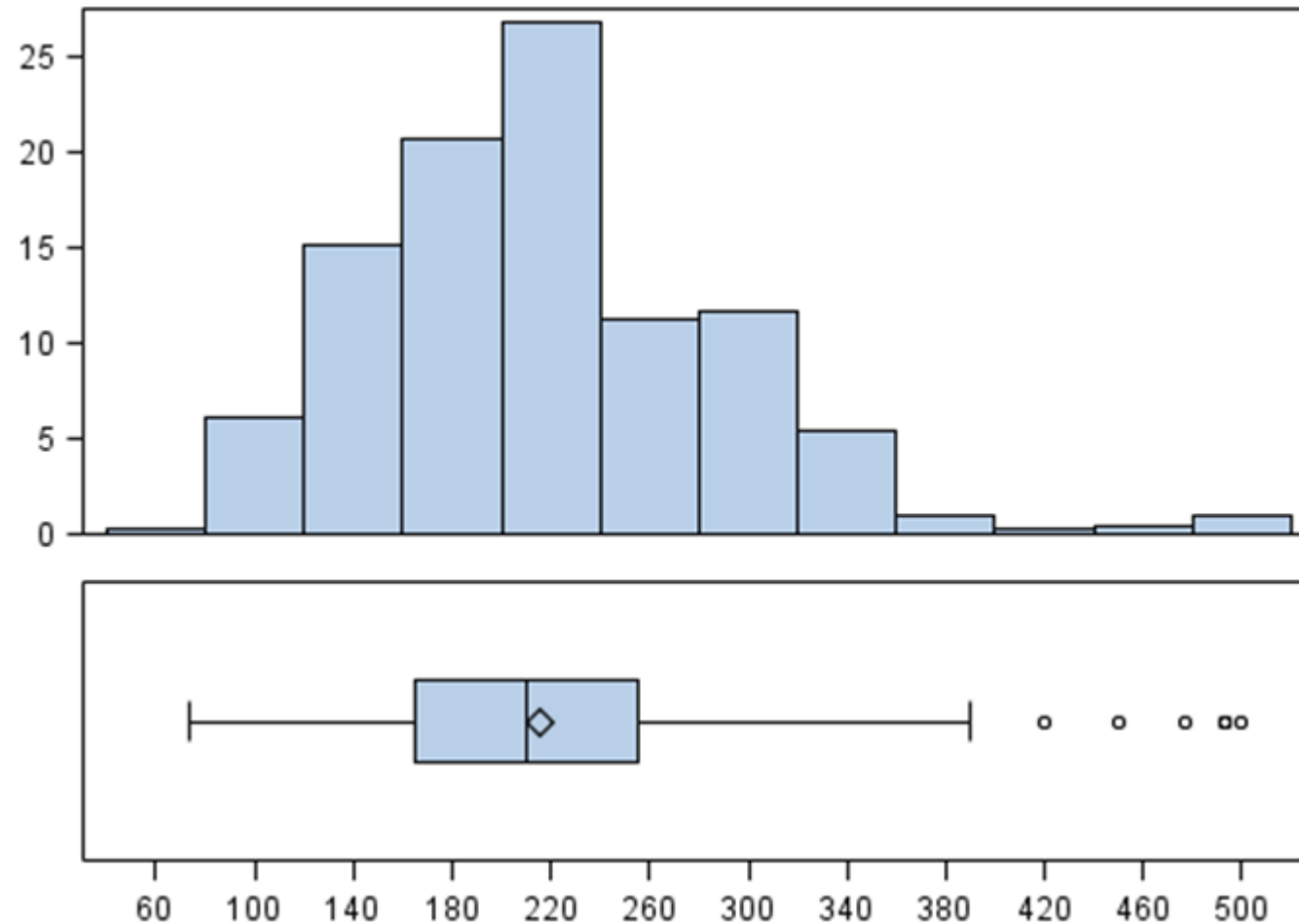
1. Drawing a box from Q_1 to Q_3
2. Dividing the box with a line (or dot) drawn at the median
3. Draw a line from each quartile to the most extreme data value that is not and outlier
4. Draw a dot/asterisk for each outlier data point.

Box plot of the number of hot dogs eaten by the men's contest winners 1980 to 2010



R: `boxplot(v)`

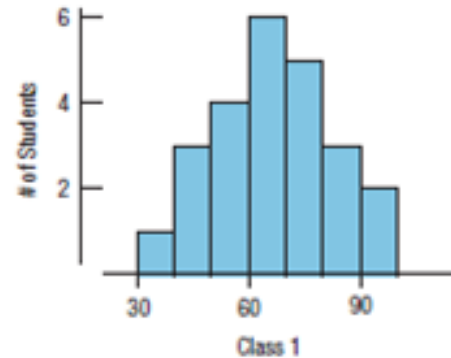
Box plots extract key statistics from histograms



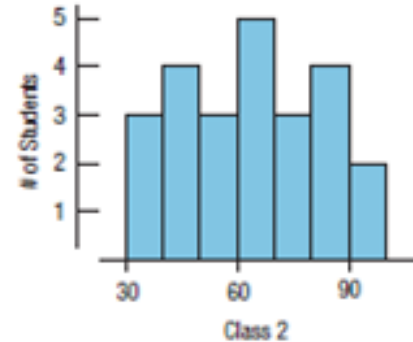
Box plots extract key statistics from histograms

Question: which Box plot goes with which histogram?

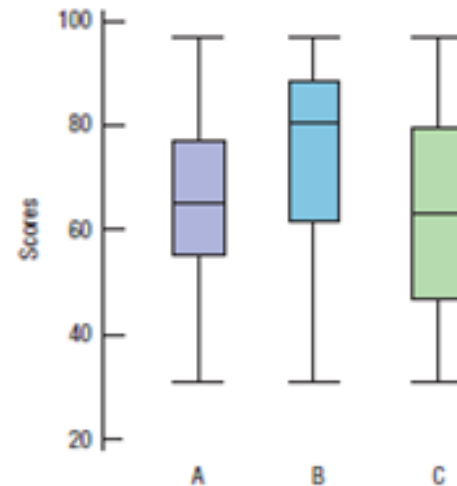
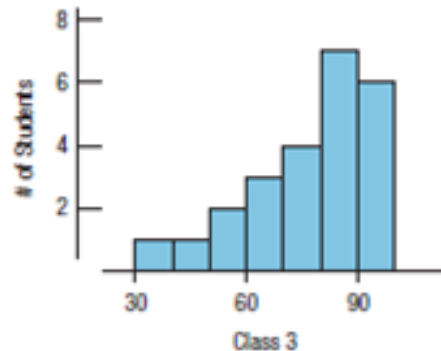
Histogram 1



Histogram 2



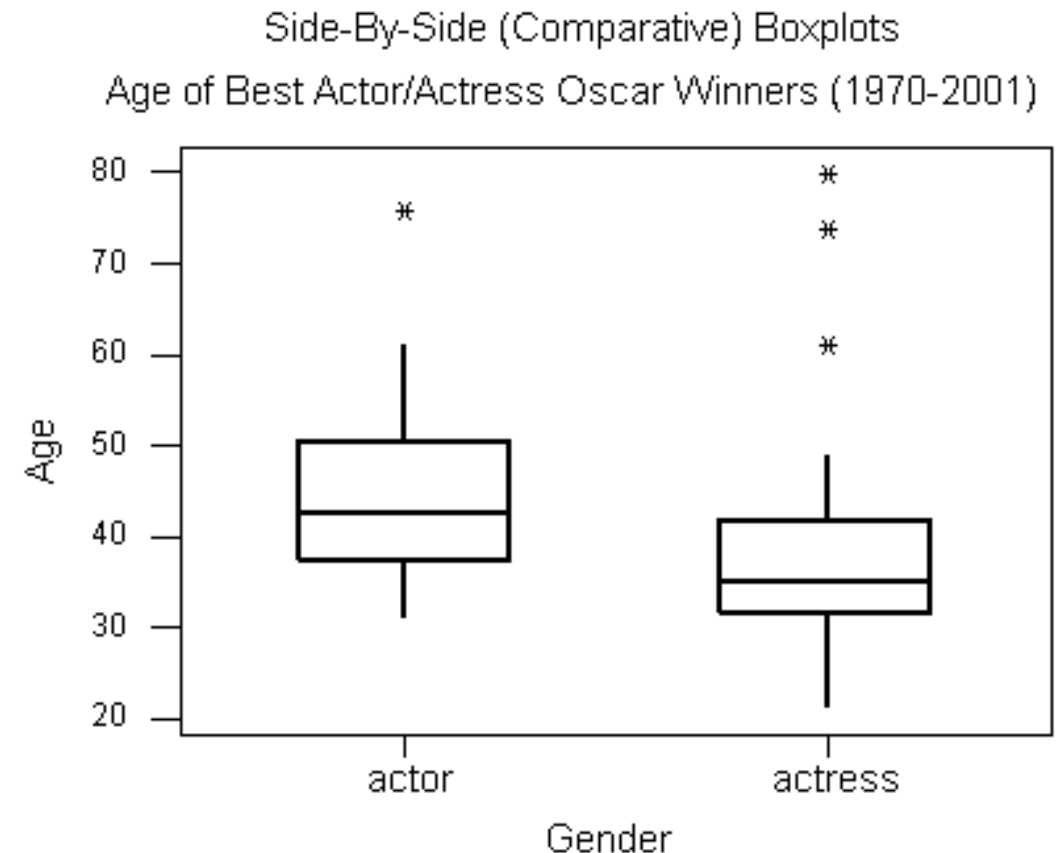
Histogram 3



Comparing quantitative variables across categories

Often one wants to compare quantitative variables across categories

Side-by-Side graphs are a way to visually compare quantitative variables across different categories



Side-by-side boxplots in R

```
boxplot(v1, v2,                                     # compare two vectors v1 and v2
        names = c("name 1", "name 2"),             # labels below each box plot
        ylab = "y-axis name"                        # y-axis label name
)
```

Let's try it in R!

Concepts for summarizing quantitative data

z-scores show how many standard deviations a point is from the mean

$$\text{z-score}(x_i) = \frac{x_i - \bar{x}}{s}$$

Quantiles show the value **x**, such that a fixed proportion of the data is less than x

- e.g., what is the value x, such that 20% of the data is less than x

Five Number Summary give key summary statistics of a data sample

- minimum, Q_1 , median, Q_3 , maximum

A **boxplot** is a visualization of the five number summary

- Side-by-side boxplots allow you to compare key summary statistics

Summary of R

We can compute a z-score for a value x , and a vector of values v using:

```
the_mean <- mean(v)  
the_sd <- sd(v)  
the_zscore <- (x - the_mean)/the_sd
```

We can compute quantiles using the `quantile()` function:

```
quantile(v, .2)    or    quantile(v, c(.25, .4))
```

We can compute a five number summary using the `fivenum()` function:

```
fivenum(v)
```

We can compute boxplots using the `boxplot()` function

```
boxplot(v)
```

Relationships between two
quantitative variables

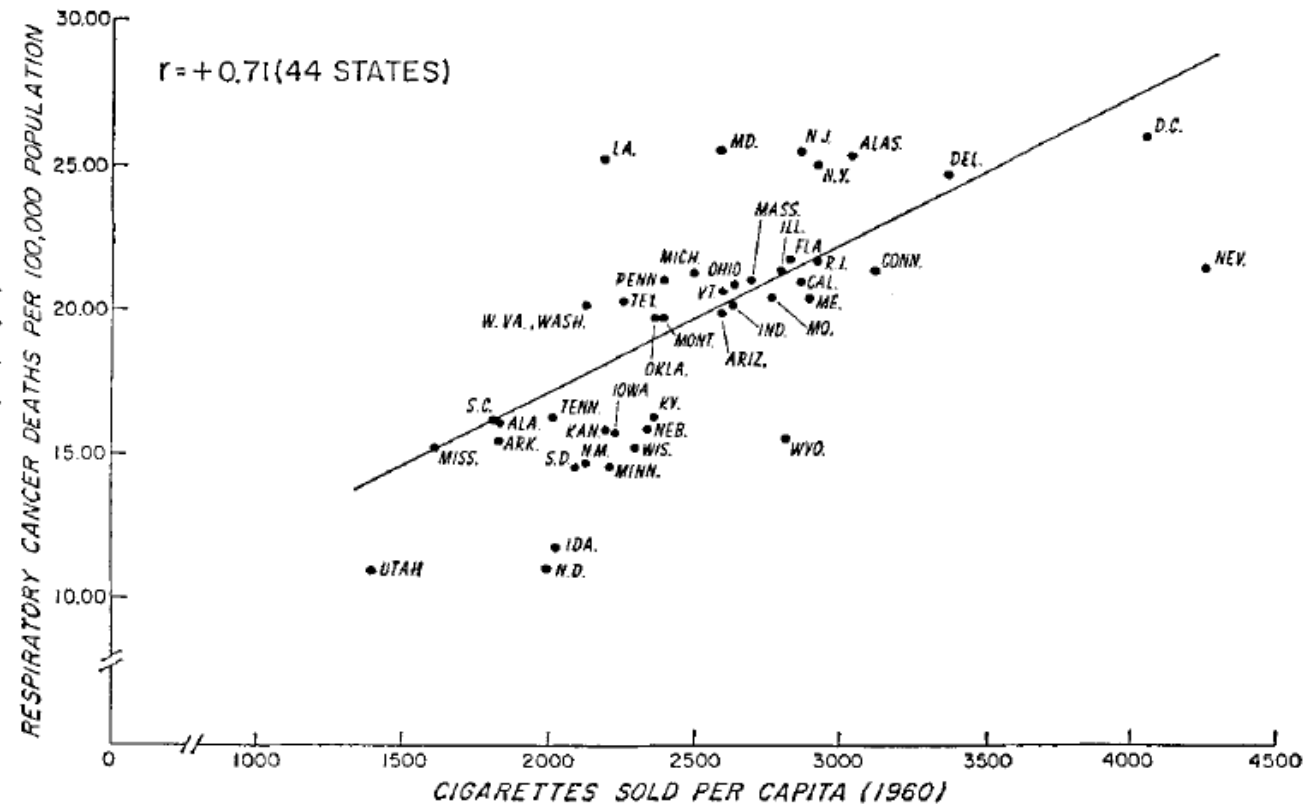
Two quantitative variables

In 1968, Joseph Fraumeni published a paper in the Journal of the National Cancer Institute that examined the relationship between smoking and different types of cancer

Explanatory variable		Response variable			
State	Cig per capita	Bladder	Lung	Kidney	Leukemia
AL	1,820	2.9	17.05	1.59	6.15
AZ	2,582	3.52	19.8	2.75	6.61
AR	1,824	2.99	15.98	2.02	6.94
CA	2,860	4.46	22.07	2.66	7.06
CT	3,110	5.11	22.83	3.35	7.2
DE	3,360	4.78	24.55	3.36	6.45
DC	40,460	5.6	27.27	3.13	7.08

Relationship between smoking and lung cancer

TEXT-FIGURE 2.—Correlation between average annual age-adjusted death rates for respiratory tract cancer (1956-61) and *per capita* cigarette sales (1960) in 44 States.



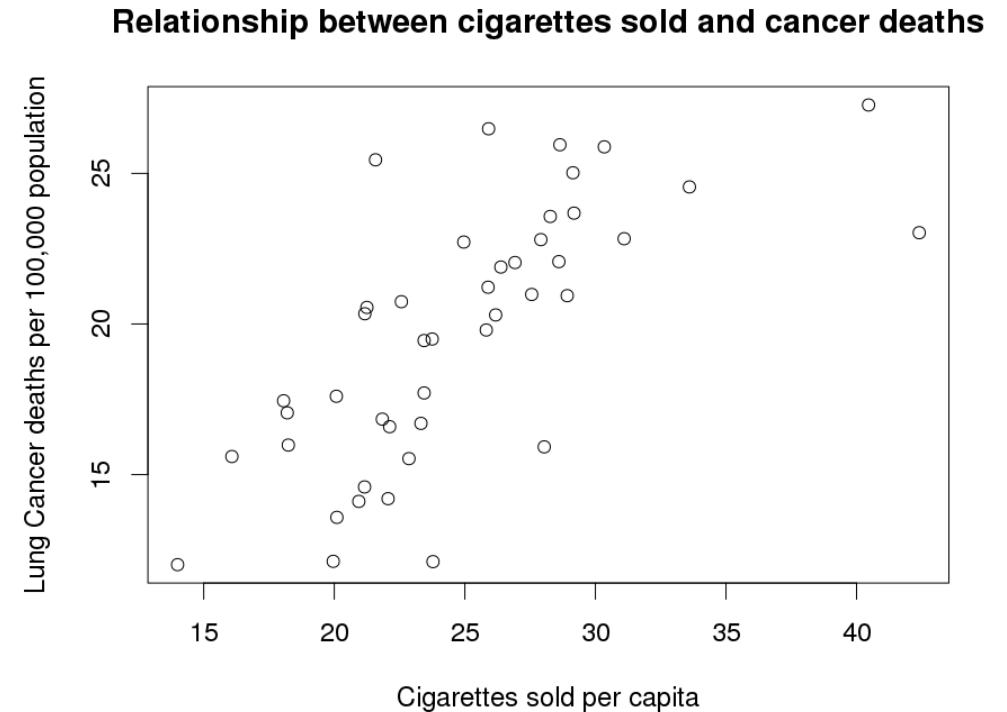
Scatterplot

A **scatterplot** graphs the relationship between two variables

Each axis represents the value of one variables

Each point shows the value for the two variables for a single data case

If there is an explanatory and response variable, then the explanatory variable is put on the x-axis and the response variable is put on the y-axis



R: `plot(x, y)`

Questions when looking at scatterplots

Do the points show a clear trend?

Does it go upward or downward?

How much scatter around the trend?

Does the trend seem be linear (follow a line) or is it curved?

Are there any outlier points?

Questions when looking at scatterplots

Do the points show a clear trend?

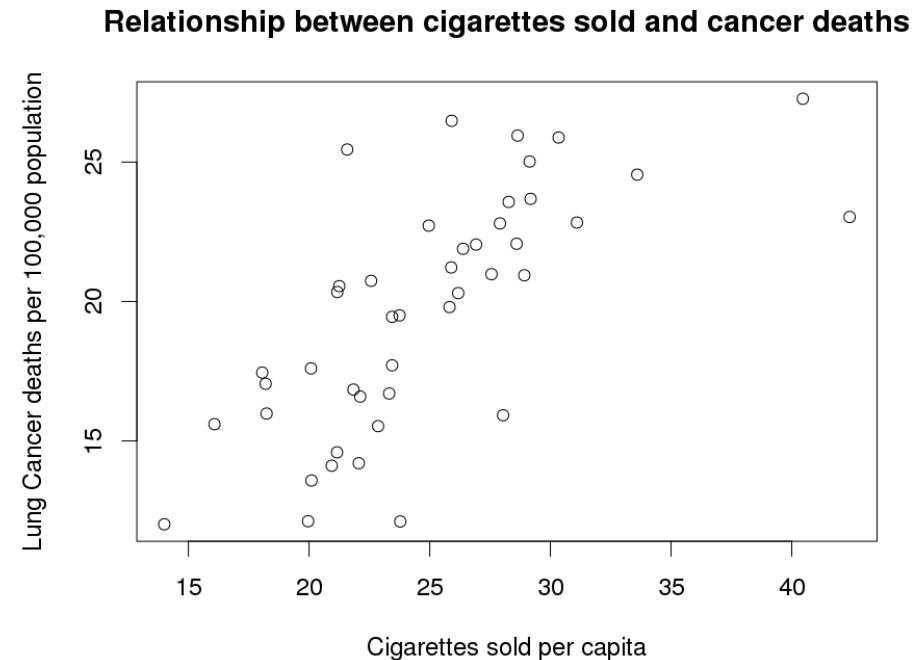
Does it go upward or downward?

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Smoking and cancer



Positive, negative, no correlation

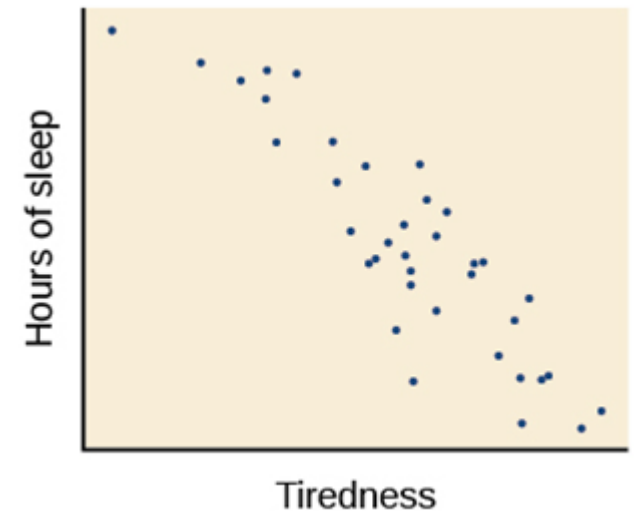
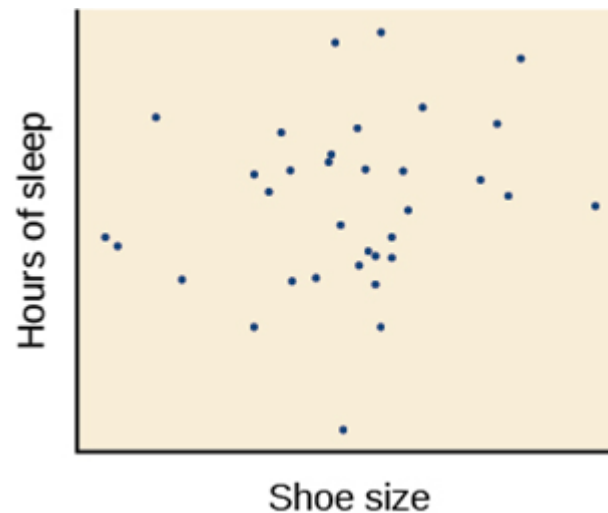
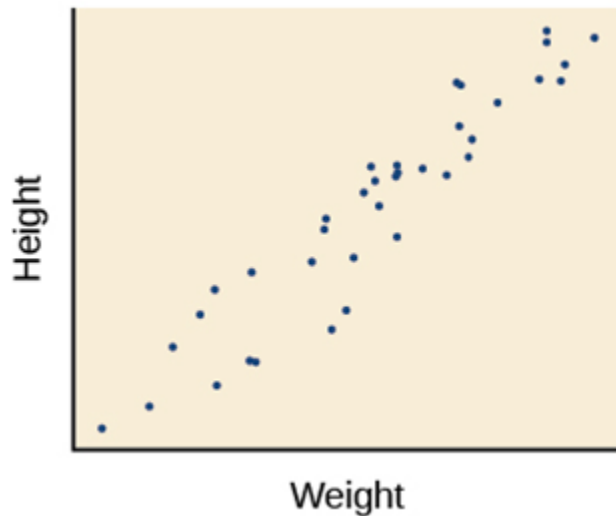
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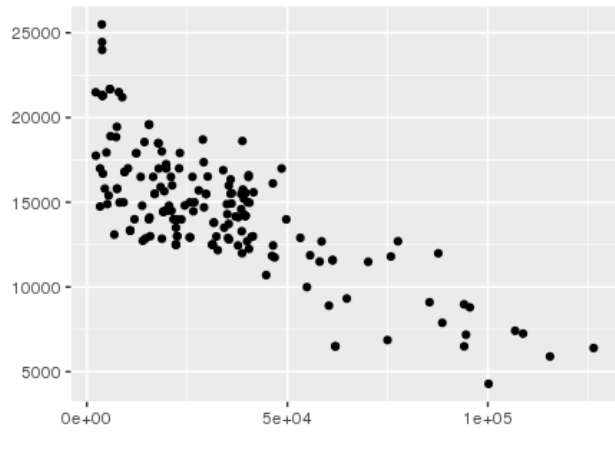
The correlation coefficient

The **correlation** is measure of the strength and direction of a linear association between two variables

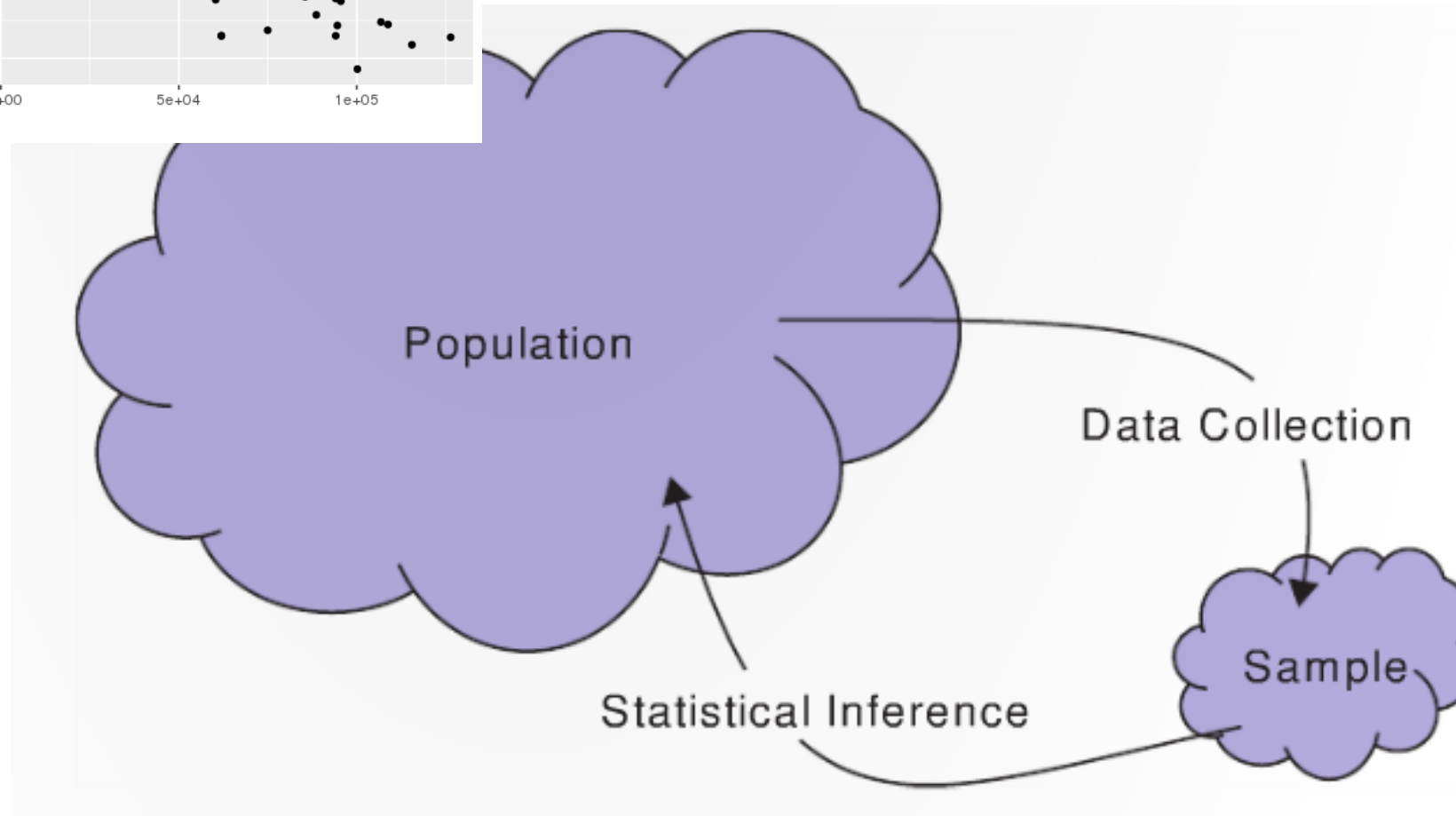
$$r = \frac{1}{(n - 1)} \sum_{i=1}^n \left(\frac{x_i - \bar{x}}{s_x} \right) \left(\frac{y_i - \bar{y}}{s_y} \right)$$

- The correlation for a sample is denoted with **r**
- The correlation in the population is denoted with **ρ**
(the Greek letter rho)

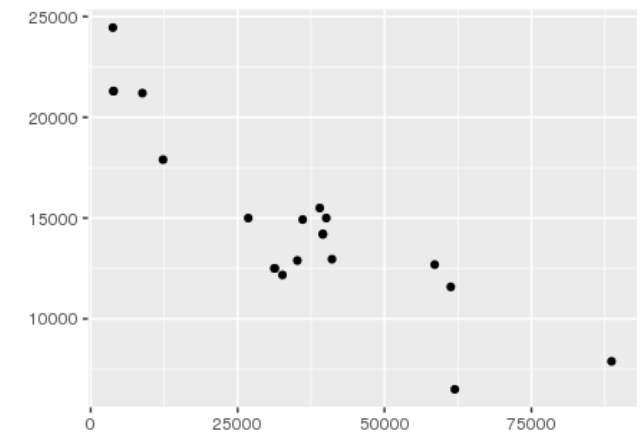
R: `cor(x, y)`



ρ parameter



r statistic

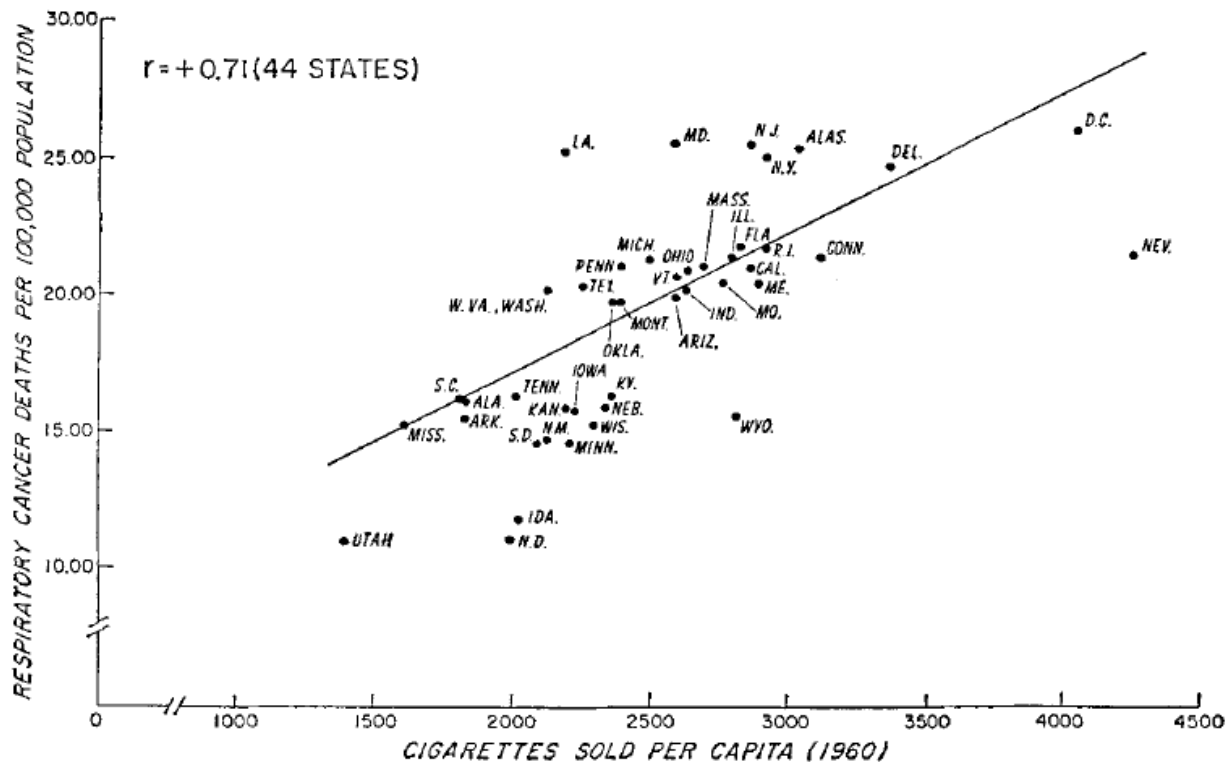


Smoking and lung cancer correlation?

The **correlation** is measure of the strength and direction of a linear association between two variables

TEXT-FIGURE 2.—Correlation between average annual age-adjusted death rates for respiratory tract cancer (1956–61) and *per capita* cigarette sales (1960) in 44 States.

$r = 0.71$



Properties of the correlation

Correlation is always between -1 and 1: $-1 \leq r \leq 1$

The sign of r indicates the direction of the association

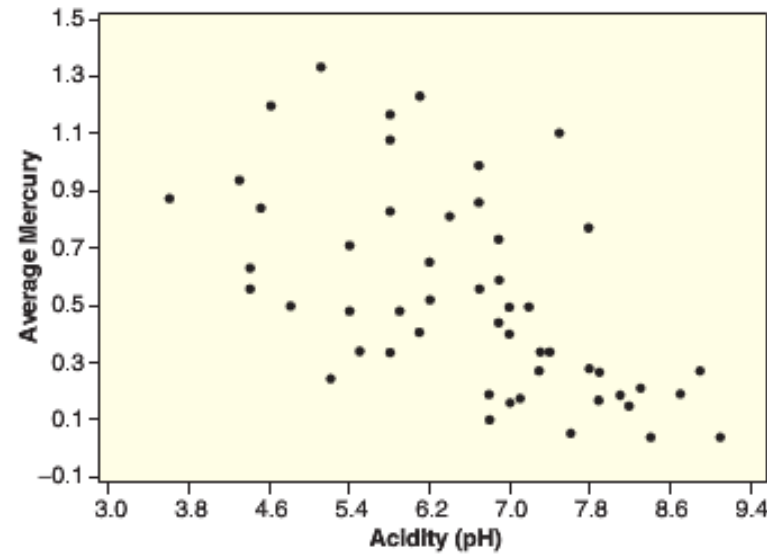
Values close to ± 1 show strong linear relationships, values close to 0 show no linear relationship

Correlation is symmetric: $r = \text{cor}(x, y) = \text{cor}(y, x)$

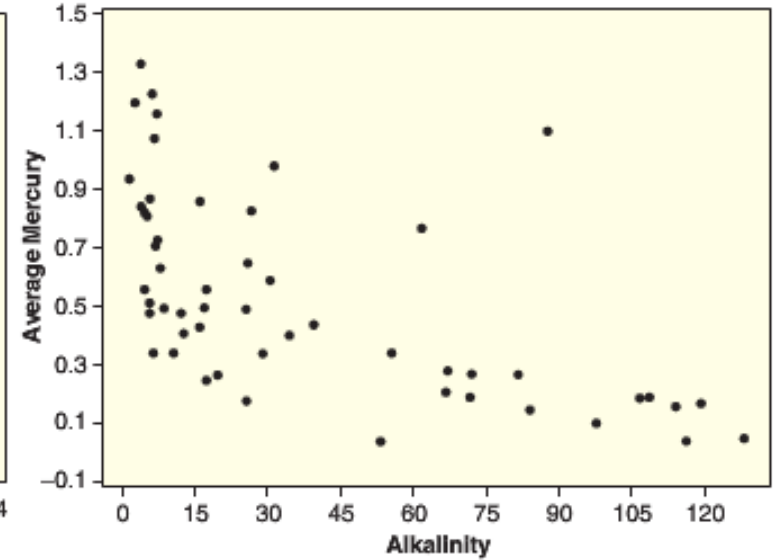
$$r = \frac{1}{(n-1)} \sum_{i=1}^n \left(\frac{x_i - \bar{x}}{s_x} \right) \left(\frac{y_i - \bar{y}}{s_y} \right)$$

Florida lakes

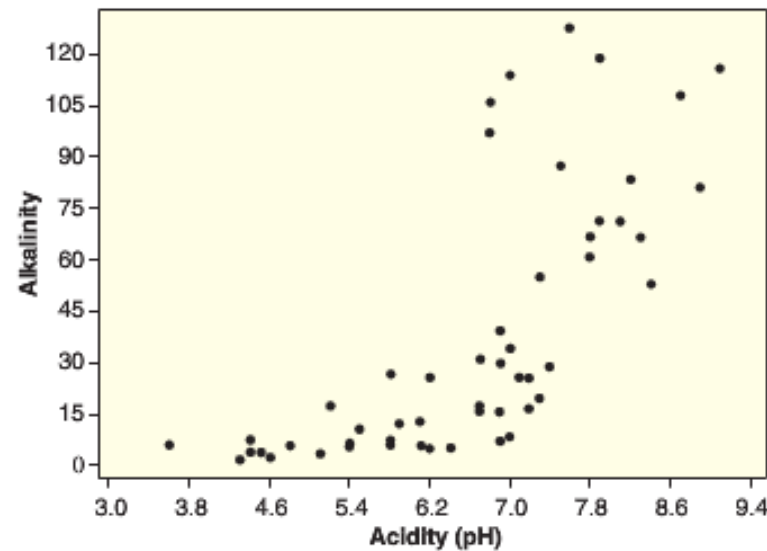
Correlation game



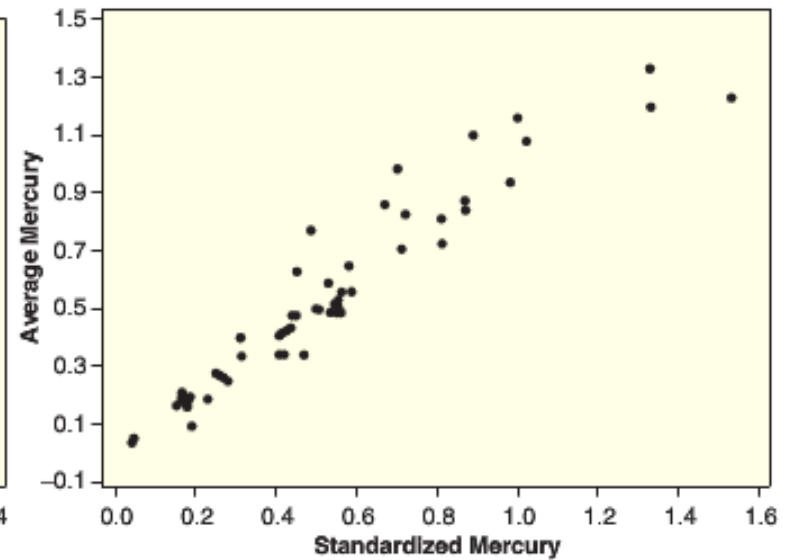
(a) Average mercury level vs acidity



(b) Average mercury level vs alkalinity



(c) Alkalinity vs acidity



(d) Average vs standardized mercury levels

Let's calculate some correlations in R

load the data

```
load("smoking_cancer.Rda")
```

create a scatter plot and calculate the correlation

```
plot(smoking$CIG, smoking$LUNG)
```

```
cor(smoking$CIG, smoking$LUNG)
```

	STATE	CIG	BLAD	LUNG	KID	LEUK
1	AL	1820	2.90	17.05	1.59	6.15
2	AZ	2582	3.52	19.80	2.75	6.61
3	AR	1824	2.99	15.98	2.02	6.94
4	CA	2860	4.46	22.07	2.66	7.06
5	CT	3110	5.11	22.83	3.35	7.20



Number of cigarette's sold per capita



Cancer rates per 10k for bladder, lung, kidney and leukemia

Try it in R

Correlation cautions

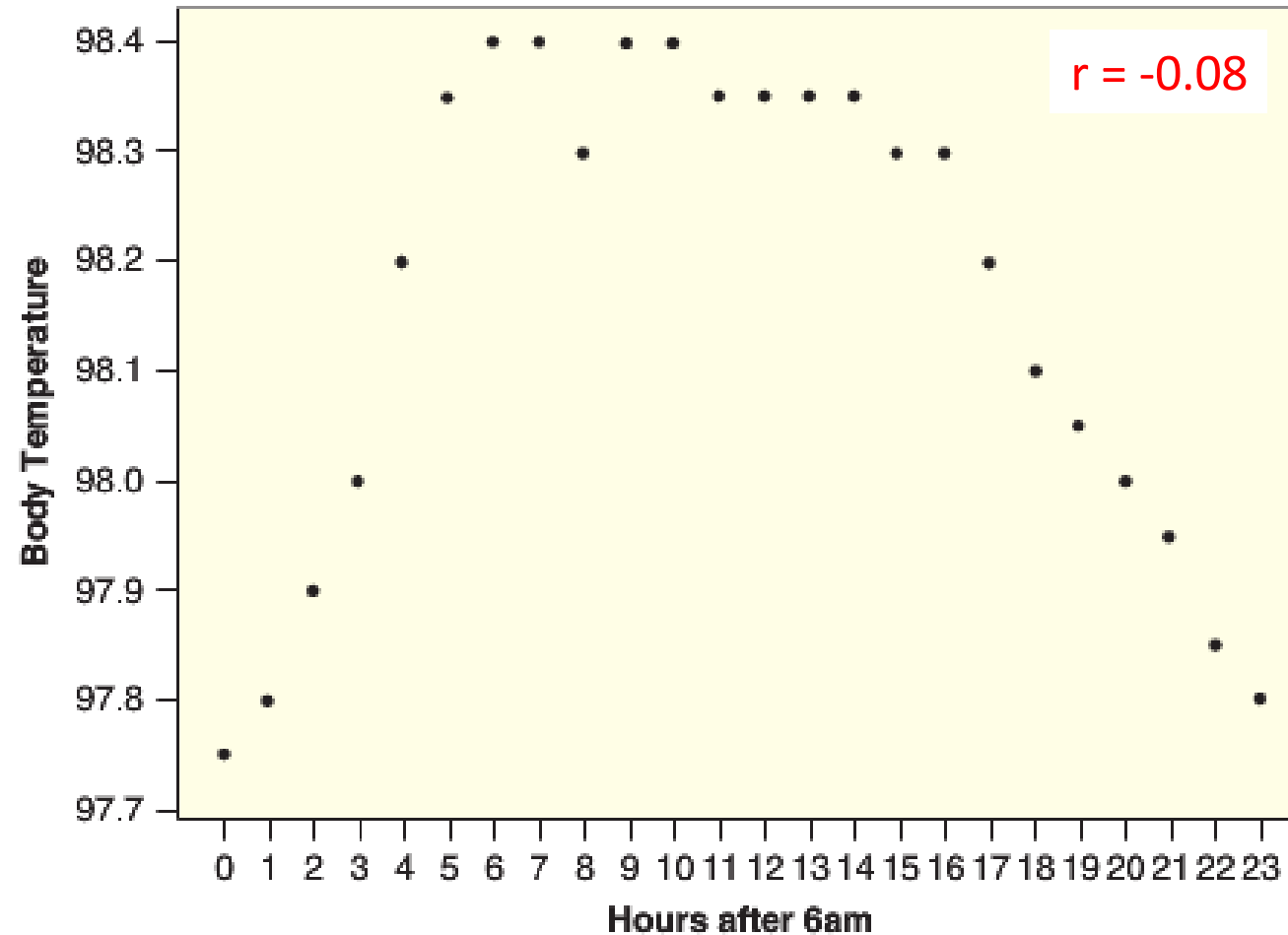
Correlation caution #1

A strong positive or negative correlation does not (necessarily) imply a cause and effect relationship between two variables

Correlation caution #2

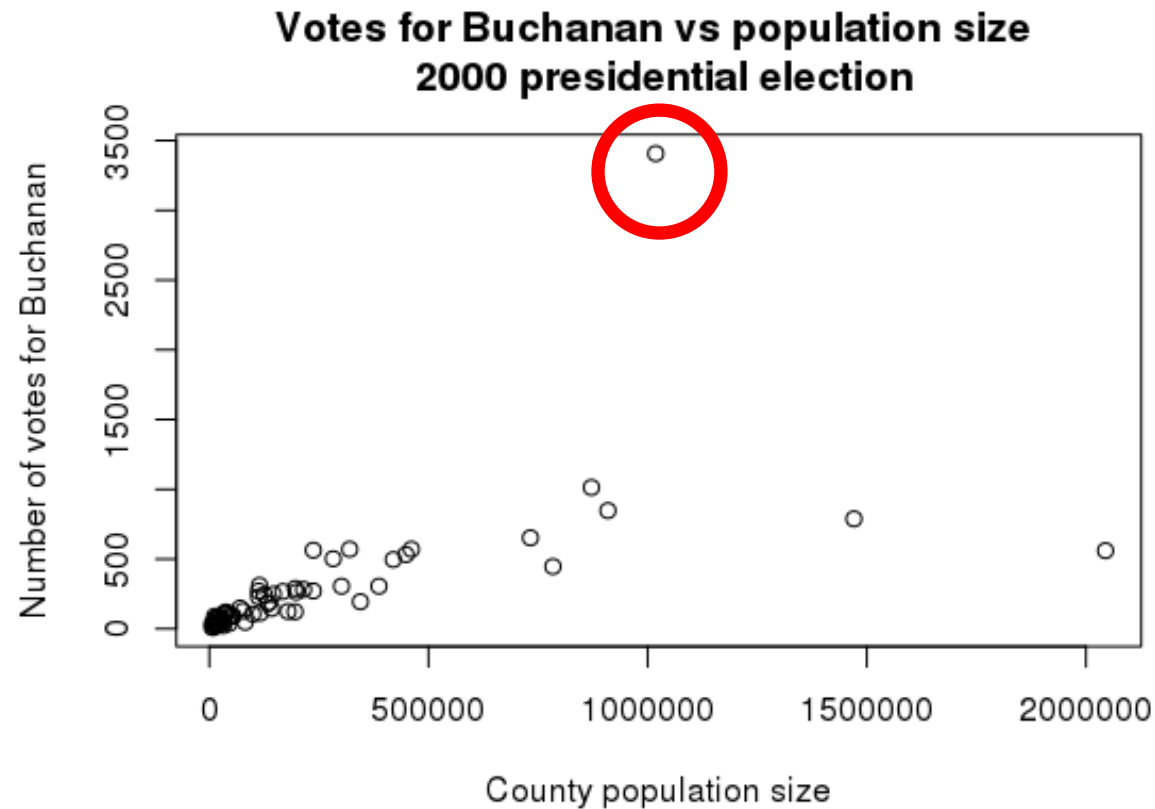
A correlation near zero does not (necessarily) mean that two variables are not associated. Correlation only measures the strength of a linear relationship

Body temperature as a function of time of the day



Correlation caution #3

Correlation can be heavily influenced by outliers. **Always plot your data!**



With Palm Beach
 $r = 0.61$

Without Palm Beach
 $r = .78$

Anscombe's quartet ($r = 0.81$)

